# Jiaheng Zheng

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## Education

**Southern University of Science and Technology** Sep. 2024 – Jul. 2028 (Expected)
*Bachelor of Optoelectronic Information Science and Engineering* *Shenzhen, China*

- GPA: 3.94/4
- Rank: 2/42
- Selected Coursework: Signal and System (99), Analog Circuit (96), Engineering Electromagnetic Field Theory (99), Solid-State Electronics (98), Advanced Mathematics (Part 2) (99)
- Other coursework I'm interested in (in progress): Stanford CS224N, CS336, MIT 6.S184, Quantum Mechanics

**University of Pennsylvania** Jan. 2027 – TBD
*Incoming Exchange Student* *Philadelphia, PA, USA*

## Research Interests

My research interests span artificial intelligence and photonics, with particular emphasis on:

- **Artificial Intelligence:** Natural language processing (NLP), multimodal agents, world models, and hallucination in AI agents.
- **Physics and Photonics:** Topological spacetime crystals, photonic integrated circuits, and optical computing.

## Programs

### Physics

**Observation of a Moving Spatiotemporal Boundary in a Transmission Line System** Sep. 2025 – Present
*Undergraduate Research Assistant, advised by Prof. Zhen Gao* *Shenzhen, China*
Artificial Microstructure Physics Research Group , SUSTech

- Developed MATLAB FDTD simulations of LC transmission lines with up to 1,600 cells to study wave propagation under temporal, traveling-wave, and separable space-time modulation.
- Computed Floquet–Bloch dispersion and weighted Zak phases to analyze bandgaps; investigated moving-interface modes using source-free eigenproblems, localization metrics, and numerical convergence checks.
- Modeled varactor circuits in Python and derived bias waveforms for spatiotemporal capacitance control, accounting for nonlinear capacitance-voltage characteristics, circuit loading, and switching transients.

**Research on Computational Optical Imaging** Nov. 2024 – Aug. 2025
*Research Assistant, advised by Prof. Jingwen Ma* *Shenzhen, China*
M-Lab, SUSTech

- Studied the fundamentals of computational optical imaging and developed practical skills in constructing simple optical setups.
- Explored the integration of optical imaging principles with advanced electronics.

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### Artificial Intelligence

**Transformer-Based Machine Translation** Aug. 2026 – Oct. 2026
*Independent Project* *Shenzhen, China*
Model Reproduction and Fine-Tuning

- Reproduced a Transformer-based model for English-to-Italian machine translation.
- Fine-tuned the model using Altair for the target translation task.

**Statistically Guaranteed Remasking for Diffusion Language Models** Sep. 2026 – Present
*Research Assistant, advised by Prof. Zhenyi Wang* *Shenzhen, China*
Conformal Prediction and Statistical Guarantees

- Developing a training-free, model-agnostic remasking strategy for masked diffusion language models, replacing CORE's hand-tuned remasking budget with a conformal threshold to target finite-sample control of the fraction of incorrect tokens left unrevised.
- Extending calibration to non-stationary streams through adaptive conformal inference, with online updates to the target miscoverage parameter and a dynamic step size guided by a distribution-shift signal.
- Evaluating the method on LLaDA-8B-Base across GSM8K, MATH, BBH, HumanEval, and MBPP against fixed-budget remasking and adaptive conformal baselines.

## Community & Leadership

**Member, APEX Honors Class of Basic Sciences** Sep. 2025 – Present
*Southern University of Science and Technology* *Shenzhen, China*

- Selected for a highly selective honors program in basic sciences (**Top 1**% of all SUSTech students).
- The program offers advanced mathematics and physics coursework, faculty mentorship, and interdisciplinary research seminars.

**Class Representative, Wolong Class** Sep. 2025 – Present
*Department of Electrical and Electronic Engineering* *Shenzhen, China*

- Selected for the Wolong Class (**Top 10**% of EEE students).
- Collaborate with faculty members across the Department of Electrical and Electronic Engineering on research activities.
- Represent classmates, coordinate communication with faculty, and organize class activities to strengthen class cohesion.

## HONORS

**First-Class Scholarship for Outstanding Students, SUSTech , 2025** September 2025

## Technical Skills

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|---|---|
| **Programming** | Python, C, Verilog |
| **AI & ML** | PyTorch, Hugging Face Transformers, NumPy, Pandas, scikit-learn |
| **Tools** | Git, Linux, Jupyter Notebook |
| **Simulation** | FDTD, COMSOL, TCAD |
| **Languages** | Mandarin (native), English (proficient; TOEFL: 5.5/110) |